

Completing a smooth affine curve and extending its group action

GPT-6.1 Sol (OpenAI)
Codex, Ultra setting

October 2026

Written and self-checked by the writing AI. No human or independent review is claimed. Original contributions are dedicated under CC0 1.0. Cited works retain their own licences.

Online course: <https://kokunoyumeto.github.io/open-math-courses/courses/AG-RG/>.

Contents

Completing a smooth affine curve and extending its group action	1
1. Finite normalization for curves	1
2. Normal local curves and the smooth completion	2
3. Extending translations as a family	4
4. How this enters the one-dimensional unipotent proof	5
Internal proof dependencies and free comparison	6

Completing a smooth affine curve and extending its group action

Original proof draft, October 2026. CC0. This lesson precedes the reductive-group lessons. Its remaining programme dependencies are listed at the end; a reference to an external proof does not discharge them.

Let k be algebraically closed. A **curve** here is an integral scheme of finite type over k of dimension one. We first construct the completion needed in the classification of one-dimensional unipotent groups. In particular, extending each individual translation is not enough: the translations must extend as an algebraic family.

1. Finite normalization for curves

We use the already proved Noether normalization theorem and its dimension formula: a finite-type domain of dimension one is finite over a polynomial subring $k[t]$. The exact earlier proof is AG-CA, *Krull dimension and Noether normalization*, Lemma 2.1, Theorem 3.1, Corollary 3.2 and Theorem 4.2, with the integral-extension proofs specified there. The theorem is used in its finite-module form, not merely as a statement about function fields.

Lemma 1.1 (a separating finite parameter). If A is a finite-type k -domain of dimension one and $K = \text{Frac}(A)$, some $u \in A$ makes A finite over $k[u]$ and $K/k(u)$ finite separable.

Proof. Choose t by Noether normalization. In characteristic zero use $u = t$. Suppose the characteristic is $p > 0$. Write $n = [K : k(t)]$. Frobenius identifies the extension $K/k(t)$ with $K^p/k(t^p)$, since k is perfect. Thus

$$[K : k(t^p)] = np, \quad [K^p : k(t^p)] = n, \quad [K : K^p] = p.$$

The tower formula gives the last equality. Some $z \in A$ lies outside K^p , because otherwise its fraction field would lie in K^p . Hence $K = K^p(z)$ and $1, z, \dots, z^{p-1}$ is a basis over K^p . A k -derivation annihilates K^p , and arbitrary assignment of its value on z defines a derivation: use the presentation $K^p[Z]/(Z^p - z^p)$. Consequently $\Omega_{K/k}$ is one-dimensional with basis dz .

The element z satisfies a monic equation

$$z^d + c_1(t)z^{d-1} + \dots + c_d(t) = 0, \quad c_i(t) \in k[t].$$

Choose a p -power P so large that $\deg c_i < iP$ for every nonzero c_i , and put $u = t^P + z$. Substitution of $z = u - t^P$ gives a polynomial equation in t over $k[u]$. Its term of highest t -degree is $(-t^P)^d$, of degree Pd and coefficient $(-1)^d$; every other term has smaller degree by the stated inequalities. Dividing by that unit makes a monic equation for t . Thus $k[t, u]$ is finite over $k[u]$. A finite $k[t]$ -module generating list for A also generates it over $k[t, u] \subset A$, so A is finite over $k[u]$.

We have $du = dz \neq 0$, hence $\Omega_{K/k(u)} = 0$. For clarity, for a finite field extension E/F the condition $\Omega_{E/F} = 0$ is equivalent to separability. A separable generator has a minimal polynomial with nonzero derivative, whose differential forces its differential to vanish; successively adjoining separable generators proves one direction. For the converse, let $F_s \subset E$ be the subfield of elements separable over F . Every element of E has a p -power in F_s : repeatedly remove zero derivatives from its irreducible polynomial. Thus E/F_s is purely inseparable. If it is nontrivial, choose a maximal proper intermediate field E' . Adjoining any element outside E' generates E by maximality. Its degree is a power p^e , and $e > 1$ would make the subfield generated by its p th power a proper intermediate field. Hence $e = 1$, and E/E' has a presentation $E'[Z]/(Z^p - a)$. Sending Z to 1 and E' to zero gives a nonzero F -derivation of E . This contradicts $\Omega_{E/F} = 0$. The choice of a maximal intermediate field is possible because the degrees are positive integers bounded by $[E : F_s]$. This proves the criterion and the lemma. $\square$

Lemma 1.2 (finite integral closure). The integral closure of A in K is a finite A -module.

Proof. Use $R = k[u]$ from Lemma 1.1 and $F = k(u)$. The trace pairing of the finite separable extension K/F is nondegenerate. Here is the needed field proof. A finite separable extension of an infinite field has a primitive element: if it is generated by two elements a, b , the finitely many embeddings into an algebraic closure have distinct pairs of images. There are only finitely many scalars c for which two embeddings take the same value on $a + cb$. Choose another c . The distinct images show that the degree of $a + cb$ equals the extension degree. Induct on the number of generators. For a primitive element v its conjugates $v_1, \dots, v_n$ are distinct. In the power basis, the trace matrix has entries $\sum_j v_j^{r+s}$, and is $V^t V$ for the square Vandermonde matrix $V = (v_j^r)$. Its determinant is the square of a nonzero Vandermonde determinant.

Choose an F -basis $b_1, \dots, b_n$ of K consisting of elements integral over R . To make a given algebraic element integral, multiply it by a common denominator of the coefficients of its monic equation; the resulting monic equation has coefficients in R . Let b_i^* be the trace-dual basis. If x is integral over R , then each xb_i and its trace are integral over R : its conjugates are integral, and the sum of integral elements is integral, by the finite-module determinant argument. But $\text{Tr}(xb_i) \in F$, and R is integrally closed. Indeed, a monic equation for a reduced fraction $a/b \in k(u)$ implies that b divides a^m , so b is a unit. Therefore

$$x = \sum_i \text{Tr}(xb_i)b_i^* \quad \text{belongs to} \quad \bigoplus_i Rb_i^*.$$

The integral closure is an R -submodule of this finite free module. Hilbert basis and the finite-submodule theorem make it finite over R , hence also over A . Since A is integral over R , integrality over A and over R are equivalent, by transitivity of integrality; transitivity itself follows by including the finitely many coefficients of integral equations in a finite R -algebra and applying the determinant argument. $\square$

Integral closure commutes with localization. One inclusion follows by localizing a monic equation. For the other, if x is integral over A_s , multiply x by a sufficiently large power of s to clear the denominators in its monic equation; that multiple is integral over A . The same argument treats a general multiplicative set by taking the product of its finitely many required denominators.

2. Normal local curves and the smooth completion

Lemma 2.1. A one-dimensional Noetherian normal local domain is a discrete valuation ring.

Proof. Let $(R, \mathfrak{m})$ be such a ring and choose $0 \neq a \in \mathfrak{m}$. Every prime containing a is $\mathfrak{m}$, since any other nonzero prime would give a chain of length at least two. Thus the radical of (a) is $\mathfrak{m}$. Finite generation of $\mathfrak{m}$ gives $\mathfrak{m}^n \subset (a)$ for some n : choose a power of each generator lying in (a) and use the pigeonhole principle on monomials. Choose the least n and $b \in \mathfrak{m}^{n-1} \setminus (a)$. Then $c = b/a \notin R$

and $c\mathfrak{m} \subset R$.

If $c\mathfrak{m} \subset \mathfrak{m}$, apply the determinant trick to multiplication by c on the finite module $\mathfrak{m}$. Its resulting monic polynomial annihilates this module, hence is zero in $\text{Frac}(R)$ because the domain has a nonzero element in $\mathfrak{m}$. Normality would imply $c \in R$, a contradiction. Consequently $c\mathfrak{m}$ is a unit for some $x \in \mathfrak{m}$. It follows that $\pi = c^{-1} = x/(cx) \in \mathfrak{m}$ and $\mathfrak{m} = \pi R$.

The intersection $N = \bigcap_j \pi^j R$ is zero. In fact N is a finite ideal, since R is Noetherian, and cancellation of $\pi \neq 0$ shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1} R$, then y lies in every $\pi^j R$. Nakayama gives $N = 0$. Repeated division of a nonzero nonunit by π must therefore stop. Every nonzero element is uniquely a unit times π^j , and every nonzero ideal is generated by the element of smallest such exponent. This is the discrete valuation ring assertion. $\square$

We use the earlier Jacobian theorem in its following proved form: a finite-type scheme over a perfect field is smooth at a point with geometrically regular local ring; in particular a normal finite-type curve over an algebraically closed field is smooth. The exact programme proofs are AG-CA, *Smooth algebras and the Jacobian criterion*, together with *Regular local rings*, including the formal-coordinate and faithful-completion arguments. To make the curve specialization explicit, at a closed point the preceding DVR has residue field k by the Nullstellensatz, and its completion is $k[[\pi]]$: successively subtract the constant residue and divide by π to obtain the unique expansion, and use $\bigcap \pi^j R = 0$ for uniqueness. In an affine presentation $k[x_1, \dots, x_n]/I$, the one-dimensional cotangent space then gives $n - 1$ equations of independent differentials. The earlier formal implicit-function proof makes their local zero set smooth of dimension one and gives completed local ring $k[[\pi]]$. The quotient to the original DVR has zero kernel: faithful completion identifies its completed map with an isomorphism, and separated completion detects a nonzero element. Thus the original curve is smooth at every closed point. The nonsmooth locus is closed by the same Jacobian minors; the Nullstellensatz makes every nonempty closed subset contain a closed point, so it is empty. This identifies the exact earlier algebra used, rather than replacing it by an external smoothness citation.

Theorem 2.2 (smooth projective completion). Every smooth connected affine curve X/k is an open subscheme of a smooth projective curve C/k , and $D = C \setminus X$ is a nonempty finite reduced set of closed points.

Proof. Smooth curves have regular local rings and hence normal local domains. Their irreducible components cannot meet, since a local domain has only one minimal prime. There are finitely many components and each is consequently open as well as closed. Connectedness gives integrality.

Choose finitely many algebra generators for X , embed it as a closed subscheme of $\mathbb{A}^n$, and take its reduced projective closure $Y \subset \mathbb{P}^n$. This is an integral projective curve and $Y \cap \mathbb{A}^n = X$. Normalize each affine chart of Y in $k(X)$. Lemma 1.2 and the localization observation glue them to a finite birational map $C \rightarrow Y$. Over X this is an isomorphism because X is normal. The resulting C is a normal integral curve, so Lemma 2.1 and the preceding smoothness argument make it smooth.

A finite map is proper: it is separated, since its affine diagonal is the closed immersion given by the surjection $B \otimes_A B \rightarrow B$; it is finite type; and it is universally closed. For the last assertion a closed subset of $\text{Spec } B$ is defined by an ideal J , and its image is $V(J \cap A)$ by lying over applied to the integral inclusion $A/(J \cap A) \subset B/J$. Finiteness, hence this argument, persists after arbitrary base change. The earlier AG-RG-S01 Lemma 7.A and Corollary 7.B therefore make C proper over k .

It is projective as well. The pullback L of $\mathcal{O}_Y(1)$ is ample: inverse images of the standard section opens are affine, since the map is finite, and their principal opens form a basis. The denominator-extension lemma proved in the earlier algebra-and-sheaves lesson extends the function defining such a principal open after multiplying by a sufficiently high power of its defining section. Thus these smaller opens too are nonvanishing loci of global sections of powers of L . The proved ample-immersion theorem then gives an immersion of C into projective space. Properness makes this immersion closed, proving projectivity. This uses the complete denominator and immersion proofs, not an unproved assertion that finite covers of projective schemes are projective.

A proper closed subset of an integral Noetherian curve has dimension zero, so has finitely many points; the reduced complement is D . It is nonempty. Indeed every rational map from a smooth curve to projective space extends at each missing point: in its DVR choose homogeneous rational coordinates, divide by the smallest valuation, and obtain coordinates in the local ring with at least one a unit. Finitely many denominators make these coordinates regular on a neighbourhood. Polynomial relations defining a closed projective target remain zero, since they are zero in the function field. Uniqueness follows from separatedness and agreement on the dense open.

If a proper integral curve had a nonconstant regular function f , its map to $\mathbb{A}^1$ followed by the open immersion into $\mathbb{P}^1$ would have closed image in $\mathbb{P}^1$ by properness. Its image contains the generic point: a nonconstant element of the one-variable function field is transcendental over the algebraically closed k . The image would be all of $\mathbb{P}^1$, contradicting its containment in $\mathbb{A}^1$. Therefore every regular function is constant. If $X = C$ were affine, the affine module-sheaf correspondence would give $C = \operatorname{Spec} k$, contradicting dimension one. $\square$

The curve C is geometrically integral. Its global functions are k by the final argument of Theorem 2.2. The proved flat base-change calculation in [Algebra and sheaf cohomology before reductive groups](#), Section 6, gives $H^0(C_L, \mathcal{O}) = L$ for every field extension L/k . Therefore C_L is connected: a nontrivial decomposition into two open-and-closed sets would give a nontrivial idempotent among its global functions. Smoothness persists after base change. Regular local domains make its finitely many irreducible components disjoint and open, so connectedness leaves one, and reducedness gives integrality. Its nonempty open X_L is integral too. Products of these geometrically integral schemes are integral: on affine charts the tensor product injects into the tensor product with the function field of either integral factor, because tensoring a vector space over k preserves injections; the latter ring is a domain by geometric integrality of the other factor. This proves the density assertions used next.

3. Extending translations as a family

Theorem 3.1. Let H/k be a smooth connected affine group of dimension one. In its smooth projective completion C , multiplication extends uniquely to a morphism $H \times C \rightarrow C$ defining an action. Every point of $D = C \setminus H$ is fixed.

Proof. Each translation on H is a birational map of C . The rational-map argument in Theorem 2.2 extends it to a morphism of C , and does the same for its inverse. Their composites agree with the identity on H , hence on C . Thus each translation is an automorphism of C .

We now prove regularity in the group variable. Multiplication gives a rational map $H \times C \dashrightarrow C$ defined on $H \times H$. Over the generic point η of H it is a birational map of the smooth projective curve $C_{k(H)}$, with inverse the generic inverse translation. The same DVR argument over that field extends it everywhere on the generic fibre. Each extension chart is given by finitely many rational functions and finitely many equations, so clearing their denominators extends it to an open neighbourhood in $H \times C$. These finitely many charts agree where they overlap: their maps to the separated target agree on the dense open $H \times H$, and the source is integral. Their union contains the whole generic fibre. The largest domain of this regular rational map is therefore an open W containing that fibre.

Since C is proper, projection sends the closed complement of W to a closed subset of H . It misses the generic point. Consequently there is a nonempty open $U \subset H$ with $U \times C \subset W$. Choose $a \in U(k)$, which exists by the Nullstellensatz. Let τ_a denote the already extended fixed translation. On the neighbourhood $V = a^{-1}U$ of the identity the formula

$$\beta(g, x) = \tau_a^{-1}(\alpha(ag, x))$$

is regular and equals multiplication on $V \times H$. For every $g_0 \in H(k)$, the formula

$$(g, x) \mapsto \tau_{g_0}(\beta(g_0^{-1}g, x))$$

is regular on $g_0V \times C$. These opens cover H : they contain every closed point, and a nonempty closed complement in a finite-type scheme over k has a closed point. They agree on overlaps by density and separatedness, so glue to $\alpha : H \times C \rightarrow C$. The unit and associativity identities hold on the dense open with all curve variables in H , and hence everywhere. Smoothness and connectedness make H , C and their products geometrically integral, so this density argument applies. Uniqueness follows the same way.

Every translation preserves H , thus its complement D . The orbit map $H \rightarrow C$ of a point of D has image in the finite set D on closed points. Its scheme-theoretic factorization through that reduced set follows from reducedness of H and density of its closed points: functions vanishing on all those points are zero by the Nullstellensatz. Connectedness makes the orbit map constant. Thus each boundary point is fixed, also infinitesimally. In particular the nonzero infinitesimal translation field on H extends to a section of $T_C(-D)$. It is nonzero because its value at the identity is the nonzero element of $\text{Lie}(H)$. $\square$

4. How this enters the one-dimensional unipotent proof

Lemma 4.1 (characters of a unipotent group). Let H be a reduced affine group of finite type over the algebraically closed k , with a closed embedding into GL_n whose closed-point matrices are unipotent. Every group homomorphism $\chi : H \rightarrow \mathbb{G}_m$ is trivial.

Proof. In characteristic $p > 0$, choose $p^e \geq n$. For a matrix $h = I + N$ with $N^n = 0$, the binomial identity gives $h^{p^e} = I + N^{p^e} = I$. Hence $\chi(h)^{p^e} = 1$. A field of characteristic p has no nonidentity root of that equation, so $\chi(h) = 1$.

In characteristic zero fix such an h and define the polynomial matrix

$$\gamma(t) = \sum_{j=0}^{n-1} \binom{t}{j} N^j.$$

For every nonnegative integer m it equals h^m . Every defining polynomial of the closed subgroup H therefore vanishes on $\gamma(t)$ at infinitely many values of t , and vanishes identically. Its determinant is one, since N is nilpotent, so this gives a morphism $\mathbb{A}^1 \rightarrow H$. The identity $\gamma(s)\gamma(t) = \gamma(s+t)$ follows from the usual binomial formula at all pairs of nonnegative integers; a polynomial in two variables vanishing on that grid is zero, by first fixing one integer variable and then the other. Thus γ is a group homomorphism $\mathbb{G}_a \rightarrow H$, and $\gamma(1) = h$. The pullback of a character is an invertible element of $k[t]$, hence a nonzero constant. The identity at zero makes it one, so again $\chi(h) = 1$.

In either characteristic $\chi - 1$ vanishes at every closed point of H . The Nullstellensatz and reducedness make that regular function zero, proving triviality as a scheme morphism. This proof uses no triangularization or solvable-group classification. $\square$

The earlier cohomology-and-models lesson proves curve duality, $\deg \omega_C = 2g - 2$, the vanishing of sections of negative-degree line bundles, and that a smooth projective genus-zero curve with a k -point is $\mathbb{P}^1$. If $g \geq 1$, the line bundle $T_C(-D)$ has degree $2 - 2g - |D| < 0$, contradicting Theorem 3.1. Hence $C = \mathbb{P}^1$.

For completeness, an automorphism of $\mathbb{P}_k^1$ is fractional linear. Its map on function fields sends t to a rational function $r(t)$ generating $k(t)$. Write $r = a(t)/b(t)$ in reduced form. The polynomial $a(T) - rb(T)$ is irreducible in $k(r)[T]$ by Gauss's lemma applied to the primitive polynomial $a(T) - Ub(T)$: a factorization over $k[T, U]$, which has degree one in U , would have a nonconstant factor depending only on T , dividing both a and b . Thus $[k(t) : k(r)] = \max(\deg a, \deg b)$. Since

that degree is one, $r = (at + b)/(ct + d)$ with $ad - bc \neq 0$. Three distinct points therefore have trivial simultaneous stabilizer; the stabilizer of one point is the affine group $t \mapsto at + b$; and the identity component of the stabilizer of two points is $\mathbb{G}_m$ after putting the points at zero and infinity. The curve action is faithful, since on its open orbit H it is left translation. With three or more boundary points the connected group's action would be trivial, an impossibility. With two boundary points it embeds in $\mathbb{G}_m$. With one it embeds in the affine group.

Here these restrictions on the automorphism family are scheme-theoretic. With three boundary points every translation on a closed point of H is the identity. The family is consequently the identity: on affine charts its coordinate differences vanish on all closed points of the reduced finite-type source, so the Nullstellensatz makes them zero. With two fixed boundary points put them at zero and infinity. On their complementary $\mathbb{G}_m$ write the pullback of t as a Laurent polynomial in t with coefficients in $k[H]$. For every closed point of H it is at ; all other coefficients vanish on all closed points and hence are zero. The inverse action makes a a unit. The action law makes $a : H \rightarrow \mathbb{G}_m$ a group homomorphism. With one boundary point put it at infinity. The pullback of t on $H \times \mathbb{A}^1$ is a polynomial in t . Each closed-point automorphism has degree one, so the same coefficient argument leaves $at + b$, with a a unit. Again a is a regular group homomorphism, not merely a character on rational points.

For a unipotent H as in Lemma 4.1, every homomorphism $H \rightarrow \mathbb{G}_m$ is trivial. Thus two boundary points would make the action trivial and are impossible. With one boundary point the slope character is trivial, so the action is $t \mapsto t + b$. The action identity says $b(gh) = b(g) + b(h)$ over every test algebra. After choosing the coordinate to give the identity of H coordinate zero, its orbit map is precisely $b : H \rightarrow \mathbb{A}^1$ and is the original open embedding $H = C \setminus D = \mathbb{A}^1$. It therefore identifies H with the additive group $\mathbb{G}_a$, as schemes and with its group law. No argument with rational points of a possibly inseparable isogeny replaces this scheme isomorphism. In particular the result applies to the one-dimensional quotients embedded in upper unitriangular groups in the later solvable-group proof.

Internal proof dependencies and free comparison

The construction above consumes the complete earlier AG-CA proofs of Noether normalization, integral extensions, the Nullstellensatz, regular local rings, and the smooth Jacobian criterion. It consumes [AG-RG-S01 Lemma 7.A and Corollary 7.B](#) for projective-space properness, projections and graphs, and Sections 8.1–8.3 of that supporting lesson for denominator extension and ample immersion. Section 4 additionally consumes the earlier curve-duality and genus-zero proofs; its character-vanishing proof is Lemma 4.1 above and does not appeal to the later solvable-group lesson. These are internal proof dependencies to be included in the programme order before use. An inaccessible former reader link is not a current accessibility receipt.

For freely readable comparison of the completion statement, see [Stacks, Tag 0H1F](#). Its citation is additional reading; Theorems 2.2 and 3.1 above provide the completion and family-extension arguments themselves.